

ISDN and IN working together

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This paper shows how IN services can be enhanced by using ISDN principles and signalling. It includes an introduction to the way in which ISDN and its supplementary services interwork with IN-based services and the associated problems which can be overcome. The main part of the paper discusses opportunities for the use of ISDN signalling to convey messaging to the user to enhance usability and to enable new and more advanced services to be offered. This includes a description of screen phones. The paper goes on to show the benefits of the combination of a strong signalling link, a screen phone and a well designed user interface on the use of services over an IN. Examples are given of how such an interface would make services intuitive to use, which would otherwise be very difficult over POTS interfaces. The paper concludes by looking towards future applications and how this type of interface and signalling architecture could be a powerful way of controlling multimedia applications.

1. Introduction

The integrated services digital network (ISDN) and the intelligent network (IN) have grown from very different beginnings. Both ISDN and IN define a whole network solution but both have been designed from opposite stand points. The ISDN standards were developed mostly from the view of the network-to-user interface, with heavily standardised services and protocols and little regard for the functions of the network elements. The IN was developed later, from the perspective of a deregulating telecommunications market-place, with companies needing to differentiate their network by providing services different from the competition. As a result, the IN standards discuss standardised service building blocks that can be put together in a wide variety of ways to allow this differentiation of services. To allow this type of service flexibility it was necessary to functionally decompose the network architecture and define the relationships and functionality of the elements within the IN architecture. As a result the IN and ISDN service offerings have grown apart, rather than together. This not only gives service problems for the customers but also represents an opportunity that has so far been missed. The ISDN has a very powerful digital bearer and signalling interface to the customer's equipment and a good network signalling system. The IN has its main advantages in rapid provision of complex services, and also has a network signalling system to match. Could the synergy of these two powerful systems be harnessed to provide a better service? This paper will provide some views on possible ways forward to exploit this opportunity and will also discuss some of the current short-term opportunities.

1.1 Signalling

The signalling systems used in IN-provided services and ISDN signalling are very different. IN-provided services are designed around the use of a plain old telephony service (POTS) telephone connection, i.e. the telephone's capabilities are simply to signal to the network using in-band tones and to be able to receive in-band tones and announcements. An example of a service operation is the use of a charge card. Firstly the user dials the number of the chargecard service; this triggers the local exchange to refer the call to the IN. At this point the IN will provide announcements to the customer instructing them to type in an account code and PIN. The customer depresses the numbers on the keypad and they are transmitted to the IN in-band. The IN receives the digits and uses them to access the customer database. More announcements and digit collection follow as the customer places the call.

Signalling in the ISDN is based upon different principles; the speech channel and the signalling information are separated into bearer (B-channel) and data (D-channel). This enables the signalling to remain separate from the speech, so that the operation of a service does not interrupt any conversation in progress. An example of this is the provision of call charge advice during a call. In an ISDN call the charge advice is requested and the information is returned in the D-channel signalling and displayed on a screen on the phone. This process leaves the conversation in

progress on the B-channel totally unaffected, so if a facsimile transmission were in progress it would continue error free. The ISDN signalling does not rely on tones, rather it uses a digital communications protocol [1]. This protocol allows for communications between the local exchange and the telephone for the provision of services. There is also provision for communications between the telephone and other service provision machines, using the service access point identifier (SAPI). This is currently not used in the network, although the use of this technique is discussed later.

Some, but not all, ISDN telephones have the capability to send in-band tones (MF4) in the B-channel, once the call is connected. This is useful in current systems to allow ISDN telephones to control IN-provided services. This is necessary because local exchanges do not pass on digits from the ISDN protocols to the IN.

2. The current position

Currently there are many detailed challenges with the interworking of the ISDN and IN; this is largely due to the different design principles behind the two sets of standards. A few examples and a view on how the industry might move forward is given in this section.

2.1 Current problems

A simple example that appeared difficult to fix was that some IN services assume a speech path is available to the customer throughout the call set-up, for announcements and digit collection. However, the ISDN does not connect the bearer channel until the called party has answered the call. This prevents the IN from collecting digits, during the call set-up for ISDN customers. To cure this problem the service control points (SCPs) have been configured to connect the call to itself for ISDN calls. This allows the provision of tones, announcements and digit collection during the operation of the IN services. This fix works for voice calls, but has drawbacks for other categories of call. This is due to the SCP connecting the call, thus preventing the process of negotiation for type of call, e.g. swapping to a data connection.

Another problem with using services provided over an IN, via ISDN access, is simply how customers can provide information to the IN. POTS customers use tones (MF4) which are automatically generated when the phone buttons are pressed. In the ISDN signalling system the equivalent of an MF4 digit is the information message (INFO), which can be generated at any point during the call and could be used to operate IN-provided services. ISDN phones could also produce MF4 tones in-band to operate IN-provided services. As a result of there being two possible methods of signalling, manufacturers' implementations are very patchy,

with some terminals producing MF4, some INFO messages and some both. Current IN systems are only designed to accept MF4 digits in-band, so a large number of terminals cannot access the services at all. Another related problem is that many of the terminals that do generate MF4 digits will only do so once the call has been connected, rather than during call set-up, when the IN may need them. Even with willing manufacturers, the terminal designs have to be updated and then the problem will only be corrected over a long period of churn in the terminal market.

Another difficult to solve problem is service interworking between IN-provided and ISDN switch-provided services. There are cases of ISDN supplementary services with similar functionality to IN-provided services. The problem here is that the two services may need to be operated in very different ways. This is confusing for the customer who may for example use the IN-provided service at home and the ISDN provided one at work. Also, if the services are invoked together, there may be undesirable interactions between them. This is only one example — there are many other ways in which service interactions between IN-provided and ISDN switch-provided services can cause problems.

2.2 The way forward

Short-term solutions to IN and ISDN interworking problems are already manifesting themselves, but these solutions often have bad side effects. These short-term fixes will continue to proliferate until the convergence of IN and ISDN is handled at a higher functional level. All through the telecommunications industry, from standards, through telcos and equipment manufacturers, there needs to be a recognition that ISDN is just another way of providing telephony services to the customer. The ISDN service has options to use data calls and is, as a result, perceived by much of the telecommunications community to be a data service. However, many customers see it as just another form of voice telephony service and so expect to be able to use all their services on ISDN as well as POTS lines. This customer-driven need means that all new telecommunications services should be designed by considering the network holistically, i.e. services must be able to be accessed over all interfaces the customer could use.

3. The future — opportunities

In the future, telcos will need to sell more advanced services to their customers. The principal reasons for this are the basic business drivers of increasing revenues and maintaining and growing their customer base. In practice, in the UK, this means providing differentiation from the competition and being able to charge for these new services. If these new and more advanced services are to be used, they must be services that the customer wants to use because

they add value to his day-to-day business. They must also be services that the customer finds easy to use, so that they will not be put off using them initially and will continue to use them regularly. The issue of ease of use is one that can be addressed by the technology, while the decision of what services the customer will want to use must rely on marketing information. This section discusses the opportunities for the future and how the power of ISDN signalling and the IN provided services might be harnessed together.

3.1 *The requirements on signalling*

It is important to have an overview of the requirements for signalling for services provided by the IN. The type and complexity of the telephone itself has a part to play in the signalling requirements. A feature-rich telephone may hold much of the service logic itself, whereas a simple terminal may need to be sent the full detail of what to display on its screen. Similarly the complexity and type of services being used will have an impact on the signalling requirements. Three examples are used here:

- charge card,
- call waiting,
- advice of charge.

These examples will help to clarify the more generic needs imposed on the signalling.

In the example of a charge card call the customer needs to be able to communicate to the network a series of numbers in a secure way, prior to dialling the destination number. The customer would like guidance in the use of the system and would ideally like to have an abbreviated method of entering the information. For example, by swiping a smart card and entering a PIN (personal identification number). The guidance as to the use of the service could be either text on a screen or a spoken instruction in-band or both. All the information for the provision of this service may be conveyed in-band without jeopardising the integrity of the service. However, where a screen-phone is in use, it would be useful to be able to give on-screen and spoken guidance at the same time.

In the example of call waiting with caller display, the network needs to send to the customer the number of the party who is calling and give an indication to the customer to tell them of the call waiting. This should ideally be done without any interruption to the call in progress. The signalling requirement can be summarised in this case as an out-of-band short message service to the terminal during a call.

Imagine a future service where the intelligence in the network is offering an advice of charge for a videophone

call. If the advice of charge information is inserted in-band the video would be corrupted, so out-of-band signalling is again required during the call.

Another major factor affecting the requirement for the signalling to support the IN is the type of interactions that the customer will have with the network. Will the customer enter a series of numbers to answer questions posed about the operation of the service or will the customer expect to use a feature-key based system? That is, a system where the terminal being used represents the functions required by a special button to invoke that service. An example to illustrate this is the use of call waiting. To invoke the service a customer with an ordinary phone has to type in a series of numbers such as * 5 7 #. However, a customer with a network services phone would have a button labelled 'invoke call waiting' that would accomplish the whole task. In the first case (stimulus signalling), the signalling system is conveying one digit at a time, probably in response to a series of announcements from the network. In the latter case (functional signalling) the signalling system can convey one message to the network which has the whole meaning. It is clear that functional signalling has the capability to make network services far easier to use and some POTS phones are now being delivered with network services programmed into some of the memory buttons. This type of facility could readily be carried one stage further — the network sending the meaning of the buttons to the telephone as and when they were needed by the customer. So a button's function and legend will vary according to the stage of a call. This adds sophistication and ease of use to the network services but requires greater signalling capacity from the network to the telephone.

3.2 *Signalling solutions*

One way of providing a signalling link from the intelligence to the telephone is to use the functional protocol [2], which is partially standardised. This protocol provides for a non-call-related signalling path based upon the register and facility messages in the ISDN D-channel. The equivalent messages in the network signalling are also partially standardised, with the relevant elements reserved for future standardisation. If this protocol were used it would provide a strong signalling pipe, with a standardised meaning, allowing telephone and software manufacturers to supply software to interwork with the provided intelligence services. There would, however, be a need to standardise an application layer protocol defining the way in which the telephones needed to respond to the intelligence commands. The downside of the functional protocol is that the pipe it provides is of limited bandwidth, being basically a (very) short message service. This bandwidth restriction limits the use of the protocol to the simple control of services and applications and would not, for example, allow the display of small still pictures on the telephone.

Another approach to the provision of this type of service over ISDN is to open a separate virtual pipe within the D-channel. This is possible using the service access point identifier (SAPI) which is intended to allow the telephone to signal to several network service processors. Currently the international standards use this method of opening up lower layer pipes to connect to the local exchange call-control and to X.25 services. Another connection could be specified which provided a connection to the telco's intelligence layer. This would effectively open a clear pipe between the microprocessor in the telephone and the intelligence in the network, allowing an efficient computing protocol to connect the two together for the provision of intelligence services. The bandwidth of this pipe is restricted to the capacity of the network to handle the data, and limited to an absolute maximum rate of 16 kbit/s on an ISDN basic rate interface.

Most forms of network provision for use of additional SAPIs groom out the traffic close to the point the access line enters the local exchange. This prevents the capacity of the links being restricted by the exchange processor capacity and moves the information on to data connections more suited to the transport of this type of message. This has implications on the network architectures for the provision of intelligence services on the ISDN and on the exchange provisioning itself, which are discussed below.

3.3 Architecture

The IN standards and architectures are explained in Abernethy and Munday [3]. This section briefly describes the implications of implementing separate signalling for services in an IN. The architectural implications of using a separate SAPI to connect the ISDN terminal to the SCP's service logic are the requirements on the local exchanges and the type of network that would be created.

Using ISDN protocols to bypass the local exchange with service control information is a powerful way of fulfilling one of the IN goals, i.e. to move the developments into the more highly reactive information technology (IT) environment and away from the traditionally slower switch suppliers. This would allow enhancements to exchanges to be limited to those necessary to support SSP functionality for POTS lines. This would also take away a major threat to the integrity of the switches, i.e. as IN-based services take off, the switch processors will gradually have more and more processing to do, until either the processors are upgraded or the exchanges are replaced.

The other major implication of grooming out these messages at the front end of the exchange is that a new overlay data network is needed for the provision of IN services. In practice this would be done by expanding and reusing current data networks. However, there would be

implications in terms of cost and operational maintenance. A significant advantage would be that the data network would be free of most of the slow-to-change telecommunications standards and it would take the whole service provision into the more reactive IT world. This would allow the upgrade of the signalling network to cope with new types of services which have not yet been thought of.

Having discussed the removal of the signalling bottlenecks, using ISDN protocols, the next section discusses how the final bottleneck — the user interface — might be overcome.

3.4 User interfaces

In order to make services over an IN easier to use, improvements to the user interface must be considered. Currently it supports only voice announcements that are responded to by strings of numbers pressed on an MF4 keypad. Many have realised that there are benefits from adding a display to the telephone, and that, by making that display large, there can be a new range of service possibilities. Indeed screen-phone market research and trials are commonplace now in the US and on the whole the trials are being well received [4]. However, the trials are mostly based upon the presentation of the information to the terminal in-band, which interrupts the speech path that may itself be in use. A simple example of this is the caller display service when used during call waiting. When a call waiting indication arrives the speech path is disconnected for roughly a second while the display information and bleep are sent in-band to the receiving customer. A shorter disconnection for the bleep only is repeated every 20 seconds or so until the incoming call is either accepted or rejected. The result of the disconnection and bleeps is to make the existing conversation more difficult to continue. This is a very useful service and the interruptions are essential for its operation. However, by transporting the information to the customer without the interruptions and bleeps, the service could be enhanced greatly. The effects of the in-band bleeps and disconnection are worse if the call is conveying facsimile or data since it will cause errors. The solution is to have an out-of-band signalling system to convey this information. The ISDN access protocol has a signalling channel (the D-channel) permanently available, out-of-band, on which this information could be conveyed. No standards explicitly exist to encourage the use of D-channel signalling for the presentation of services over an IN. However, it is a very powerful method especially when combined with screen-phone terminals to display prompts to the customer. Currently IN services are based upon the use of voice announcements asking the user to press numbers on a telephone keypad which pass in-band tones. The sequences of announcements and responses can become difficult to use. A simple example is given in Fig 1

of the retrieval of a voice message from a network-based call answering service.

- 0800 xxxxxx
- ⇒ Welcome to the 'Mobility co' personal mobility service
- ⇒ Please enter your pin number followed by a hash
- xxxx#
- ⇒ You have one new voice message
- ⇒ Press one to make an outgoing call
- ⇒ Press two to hear your voice messages
- ⇒ Press three to configure your services
- 2
- ⇒ You have one new voice message
- ⇒ <voice message played>

Fig 1 An example of message retrieval.

This interface is usable, but it has to be regarded as less than ideal. There is a distinct limit to how many messages can be listened to before the customer starts to forget the earlier options. This is creating a bottleneck between the intelligence in the network and the customer, which, if removed, would allow far easier use of more complex services. A stronger link would allow more information on the use of the service to be passed to the customer and the customer to pass more information back to the service logic of the intelligence. This in turn would help to make the services far easier to use and would result in the development of far more powerful, but still easy to use, services. Figures 2 and 3 show how the same service could be presented by using a screen-phone interface. The screen-phone consists of a handset and a large display which is touch sensitive. This means the network can program buttons to be anywhere on the screen, with text or graphics in them, and the user simply touches the button on the screen. This gives an easy to use access medium for the integrated display of both information and IN-based services. Compare the example in Fig 1 to the use of a screen-phone interface for this type of service, as shown in Figs 2 and 3.

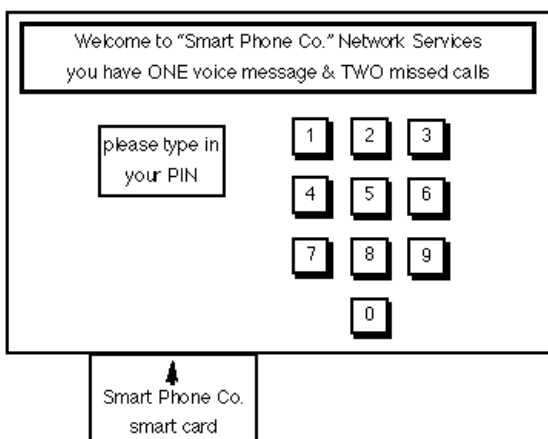


Fig 2 Screen-phone — insert smart card and type in PIN.



Fig 3 Screen-phone — select whichever option required.

The diagrams of the screen phone in Figs 2 and 3 are self-explanatory, showing how easy it could be to use an interface of this type. Simply inserting a smart card and typing in a PIN gives access from any suitable phone to the customer's own services. Then, pressing one button selects the service and allows the voice messages to be played. This type of screen interface could provide an integrated, easy-to-use access to all IN-based services and could also provide an excellent window into what are currently known as information services, such as local information pages with train and bus times, booking of cinema tickets, home shopping, home banking, etc. By using a good-sized screen on a phone, the interface to such services can be very powerful and very easy to use. Imagine seeing a plan of the cinema, on screen, before choosing which tickets to buy, or a full bank statement on the screen with the ability to look in more detail at particular transactions on-line. Many other information services are possible and would be available in a simple, easy-to-use format without having to own and understand a personal computer (PC). The result is that these services would be available to many more people than the small percentage of home computer users. This increased market penetration could enable this type of service to take off quickly. A relatively low-cost screen-phone could get the information revolution into homes and businesses very quickly.

3.5 The benefits of screen-phones

To further illustrate the benefits of using a screen-based system with an advanced network intelligence, compare the following two short stories both about the same working day a few years in the future.

- Simon is a marketing executive for a big company. Each day he gets up at 6.30 so that he can have breakfast before he drives to the train station to catch

the 7.36 train. He arrives in the office at about 8.25. Simon has recently bought a mobile phone so that he can keep in touch while he is on the train and has subscribed to some of the new services from A.Telecom which allow him to divert his phone calls from his home phone to the mobile. His work PBX allows him to forward his calls from the office to his mobile as well, so Simon feels that he is now always in touch with his customers.

It's Tuesday morning, 6.20, Simon wakes abruptly to the sound of a bell ringing, he reaches for the alarm clock to turn it off, the bell continues. Startled and suddenly wide awake Simon struggles with the alarm clock for a second or two before reaching across to pick up his mobile phone and answer the call. "Oh! Hi Simon, it's Larry, I didn't expect you to be at work at this time, I was just going to leave a message on your machine." "Argh" replied Simon, "I've got my office phone diverted to my mobile number." "I suppose you thought that was clever," retorted Larry; "I was just ringing to let you know that Barbara has just gone into labour, I'm at the Hospital now, so I won't be in today — is that OK?" "Oh yes, fine, good luck, see you in a few days I guess?" "Thanks, yes err.. bye then." Simon puts the phone down, and collapses back on to the bed. Three minutes later a bell rings again. Simon reaches out of bed and throws the mobile phone into the linen basket, the bell continues to ring, but now much louder; startled and suddenly awake Simon turns off the alarm clock, gets out of bed and retrieves the mobile phone. Later Simon picks up his mobile and as he leaves the house he types in the 18 digits from his crib sheet that diverts his home phone to his mobile. He drives to the station and gets on to his usual carriage of the train. First thing today Simon has an important meeting, so he wants to read up the paperwork on the train. The papers are still in his office because he forgot to bring them, so he reads the sports pages as usual. He is not particularly interested, but it passes the time. On arrival at his office Simon finds the lifts are out of order so he has to walk up to the 9th floor meeting room. He is a little late so is surprised to find nobody there. After a few minutes Simon becomes concerned, so he phones his office to find out what is going on. His mobile phone rings. "Typical" he mutters slamming down the phone. He picks up his mobile but as he does so it stops ringing. The penny drops. Simon dials the front reception and is put through to his office, where Mary tells him that the meeting has been cancelled. "Didn't you get the message, I left it on the agenda on your desk?" Simon slams the phone down and walks down the 7 flights of stairs to his office. When he gets there he finds Mary's message on top of the piles of papers he left on his desk the night before. "You look a bit out of breath" quips Barry; "Donald from A&B leisure came in a couple of minutes ago, wanted a quote so I

took care of it — lucky I was here otherwise we would have lost his business." It had always annoyed Simon that Barry always seemed to be in the know about everything and this just reinforced his annoyance. Barry is the same grade and works on the same type of accounts as Simon, but seemed somehow to be doing a bit better at it.

Barry and Simon's lives were actually quite similar, both living a similar distance from work and travelling in by train.

- Barry also got up at about 6.30. This morning when he got up he looked at his screen-phone in the bedroom and noticed that he had a message. He pressed the message button and saw that it was from Larry. It was timed at 6.18 which he thought was strange, but he noted that it had come in as a business call so he knew he could handle it later. Over breakfast Barry always checked his paper mail, and usually had a quick look at his electronic mail using the screen-phone on the breakfast bar in his kitchen. He pressed the e-mail button and saw a note from Mary entitled 'meeting postponed'. He scanned through it on the screen and noted that the early morning meeting had been postponed until the afternoon. He tapped a couple of soft keys on the screen to store the message as a reminder to re-arrange his afternoon meeting. He looked at another electronic message from his credit card company, this time a bill. He read down the list, on screen, of things he had bought, before authorising payment of the bill from his current account. He finished his breakfast and on his way out to the car picked up his personal digital assistant (PDA) from its charger and put it in his pocket. The charger notified the fact that he was now mobile with the network intelligence service from B.Telecom. According to his personal profile, set up with B. Telecom service, all calls were now automatically directed to a voicebank service for 10 minutes while he drove to the station and parked the car. Once on the train Barry felt happy to take calls again so the service directed all the calls to his PDA, which had a digital mobile phone built in to it. Barry sat down at his usual table and went through his usual routine of checking his daily appointments and messages that had been downloaded to his PDA. He checked the voice message from Larry and sent an e-mail reply from his PDA keyboard wishing him good luck with the birth. He then went on to re-arrange his afternoon meeting, using the PDA's capabilities to log on to the office network. While he was logged on he downloaded the summary of the papers for the morning meeting which Simon had forgotten to copy to him. Barry spent the rest of the journey reading the summaries and making a few notes for the meetings.

As Barry climbed the last few stairs towards the office he saw Simon running up past the office. He shouted out to tell him that the meeting had been postponed, but Simon was already out of earshot.

This story highlights a few examples of what a well integrated personal mobility service on an intelligent network with a screen-phone and PDA interface can do for customers. Some of the facilities might sound a little far fetched, but all could easily be possible with the enabling force of a screen-based interface to an intelligent network and sufficient signalling power to enable the transfer of the data. Indeed some features not mentioned in the story, such as home shopping, would be provided complete with good quality pictures of the items being bought.

4. A first step into multimedia and information

It has already been illustrated that screen-phones form a first step into multimedia and information services. In their most basic form they allow simplified explanations of the intelligent network services and text-based access to information services, such as bus and train timetables and simple home shopping. This in itself is a first step into the information arena, but, with the greater bandwidth available from ISDN signalling, such screen-phones can offer graphical representations, pictures and even video calls. A typical application might be home shopping, where text and graphical menus allow quick and easy selection of products, which can then be viewed as still pictures or a short video clip. Another service might allow a look-up of train times and delays, on screen, while arranging a meeting over the voice channel. All of these types of services are possible and form a valuable first step into the information revolution. Screen-phone customers will be able to use all their advanced services without having to be tied to a PC or a VOD terminal. Many of these services can be provided via a PDA [5]. Information services such as those described could be provided over many different types of equipment, with differing capabilities. In an office, services could be accessed from a PC, at home via a screen-phone or a VOD terminal, and while on the move via a PDA. Each type of terminal will have its place in a portfolio of products to enable access to information services. Much of the technology is available now and could provide an early introduction of information services. This is possible using the narrowband network, without waiting for broadband technology to take off.

5. Conclusions

This paper has shown that the combination of the power of the signalling and data capabilities of the ISDN network and the services provided over an IN are complementary. It should be seen as a powerful and lasting relationship, bringing easy-to-use intelligent network and information services to all customers. This combination of ISDN and IN technology would allow the integrated presentation of advanced intelligent network and information services without waiting for the take-off of broadband services. This is a technology that could be put into place in a short period and would be a first and relatively inexpensive step into information and multimedia services. This would, in itself, open up a change in culture and help to pave the way into full interactive broadband information and multimedia services.

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